

Constrained Optimisation

LLE – Mathematics and Statistics Skills

1. You have the choice of two products, product x and product y. Product x has a cost of £4 and product y has a cost of £1. You have a maximum of £60 that can be used for purchases.

The amount of utility is given by the equation:

$$U(x, y) = x^2y$$

- (a) If you bought 8 units of product x, what is the maximum number of units of product y you could buy?
 - (b) What would be the utility gained from buying the products in part a?
 - (c) How much utility would be gained from buying 12 units of product x and the maximum number of product y?
 - (d) Write down the constraint function.
 - (e) Use a Lagrangian to find the optimal number of each unit that should be bought and the utility this brings.
 - (f) Do your answers to b, c, and e suggest this is a maximum or minimum solution?
2. Someone decides that the amount of 'brain readiness' (B) from preparing for an exam via reading (r) and via exam practice (p) is given by:

$$B = r^2p^3$$

Given that there is a maximum of 8 hours of time each day to prepare, how much time would you give to reading and to exam practice each day?

3. A utility function is given by $U(x, y) = x^{0.6}y^{0.4}$. The cost of good x is £5 and good y is £8. The budget is £1200. Find the optimal number of each good to buy and the optimal utility.
4. For the utility function of question 3, a utility of 75 must be achieved. The cost of goods x and y are £5 and £8.
- (a) Write down the cost function C .
 - (b) Write down the constraint function.
 - (c) Write down the Lagrangian.
 - (d) Find the minimum cost and the amount of the two goods that should be bought.
5. My blood sugar (S) is made up of chocolate (c) and fudge (f). It can be calculated with the formula:

$$S = c^{0.25}f^{0.75}$$

It is important that my blood sugar is 100 units during each day. The cost of 1 chocolate is 2p and the cost of 1 fudge is 8p. How much of each should I buy so that I minimise the cost?